

Campylobacter Report: Results

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March 17, 2026

1 Behav

1.1 Tables

Table 1: Descriptive statistics of child behavior during structured observations at two timepoints.

	Timepoint 1 (May–Dec 2021)	Timepoint 2 (Dec 2021–June 2022)
# subjects	158	152
mean age (range) in days	184 (132–263)	371 (338–466)
duration of observation (hours)	698	608
# duration-based activity transitions	3044	2517
average % of time spent awake	63.6	77.1
average % of time spent bathing	0.2	0.2
average % of time spent drinking	14.3	10.5
average % of time spent sleeping	21.9	12.1
# frequency-based activities	23677	30801
rate of eating (times per hour)	0.36	2.20
rate of mouthing (times per hour)	13.27	10.68
rate of pica (times per hour)	0.05	0.67
rate of touching (times per hour)	20.28	37.08
# compartment transitions	2753	3780
average % of time spent carried by mother	40.3	26.0
average % of time spent carried by other	14.5	6.9
average % of time spent on the bare ground	0.2	16.6
average % of time spent on a surface with barriers	45.0	50.5
# location transitions	44	150
average % of time spent out of homestead	0.4	2.2
average % of time spent within homestead	99.6	97.8

Notes: Statistics were calculated from the cleaned long-format behavior datasets for the two timepoints. Duration-based percentages were summarized from activity, compartment, and location records. Frequency-based rates were computed as event counts per hour of observation.

2 Environment

2.1 Tables

Table 2: Number of samples and censoring proportion by environmental source.

Source code	Source	Total samples	Censored samples	Proportion censored
EB	Bathing water	44	14	0.32
ED	Drinking water	148	76	0.51
EF	Food	96	87	0.91
ES	Soil	148	48	0.32
EV	Fomite	265	234	0.88
HA	Areola swab	151	150	0.99
HB	Breast milk	143	143	1.00
HH	Sibling hand rinse	125	69	0.55
HP	Mother hand rinse	152	94	0.62
HR	Infant hand rinse	152	121	0.80

Table 3: Posterior summaries of the censored normal model parameters (μ and σ) for each environmental source (all-data).

Source code	Source	Parameter	Mean	SD	2.5%	50%	97.5%	n_{eff}
EB	Bathing water	μ	3.68	0.23	3.22	3.68	4.10	2312
		σ	1.39	0.20	1.05	1.37	1.86	2576
ED	Drinking water	μ	3.11	0.13	2.84	3.11	3.35	1415
		σ	1.21	0.12	1.01	1.20	1.47	1379
EF	Food	μ	-0.51	1.58	-4.35	-0.19	1.25	297
		σ	2.62	1.01	1.48	2.41	5.23	289
ES	Soil	μ	3.97	0.08	3.81	3.98	4.13	1945
		σ	0.87	0.07	0.75	0.87	1.01	2343
EV	Fomite	μ	0.65	0.50	-0.45	0.71	1.47	827
		σ	2.15	0.34	1.60	2.11	2.94	801
HA	Areola swab	μ	3.00	0.15	2.69	3.01	3.28	1631
		σ	1.30	0.14	1.05	1.29	1.60	1655
HH	Sibling hand rinse	μ	2.74	0.17	2.36	2.75	3.04	1440
		σ	1.42	0.16	1.15	1.40	1.77	1296
HP	Mother hand rinse	μ	1.95	0.31	1.20	1.99	2.44	718
		σ	1.45	0.24	1.07	1.42	2.01	717
HR	Infant hand rinse	μ	2.74	0.17	2.36	2.75	3.04	1440
		σ	1.42	0.16	1.15	1.40	1.77	1296

Notes: HB (breast milk) only has censored data, so no data for fitting.

Table 4: Posterior summaries of the censored normal model parameters (μ and σ) for each environmental source (by Timepoints).

Source	Timepoint	μ	σ	$\text{Var}(\mu)$	$\text{Var}(\sigma)$	$\text{Cov}(\mu, \sigma)$
EB	T1	3.683	1.026	0.051	0.045	-0.006
	T2	3.515	1.944	0.290	0.297	-0.117
ED	T1	3.100	1.338	0.044	0.038	-0.019
	T2	3.123	1.096	0.028	0.024	-0.012
EF	T1	-0.291	2.448	10.964	5.624	-7.285
	T2	-0.945	2.952	5.261	2.175	-3.168
ES	T1	4.385	0.665	0.007	0.005	-0.001
	T2	3.634	0.863	0.013	0.010	-0.003
EV	T1	-0.131	2.402	1.651	0.655	-0.977
	T2	1.024	2.063	0.347	0.184	-0.225
HA	T1	NA	NA	NA	NA	NA
	T2	-0.103	1.472	22.301	4.518	-9.720
HB	T1	NA	NA	NA	NA	NA
	T2	NA	NA	NA	NA	NA
HH	T1	2.369	1.588	0.129	0.113	-0.084
	T2	3.452	1.063	0.026	0.024	-0.009
HP	T1	2.194	1.374	0.122	0.087	-0.080
	T2	3.236	1.332	0.038	0.033	-0.015
HR	T1	-0.807	2.357	6.708	2.482	-3.701
	T2	2.816	1.113	0.043	0.034	-0.024

Table 5: Posterior summaries of regression coefficients in the censored normal regression model with sample source as a covariate. The baseline category is bathing water (EB).

Parameter	Source term	Mean	SD	2.5%	50%	97.5%	n_{eff}	$\hat{R}$
β_1	Intercept (EB)	3.733	0.188	3.371	3.731	4.113	728.96	1.006
β_2	ED	-0.613	0.219	-1.050	-0.611	-0.196	811.12	1.005
β_3	ES	0.157	0.214	-0.275	0.160	0.563	834.65	1.005
β_4	HH	-0.672	0.223	-1.110	-0.671	-0.243	827.66	1.007
β_5	HP	-0.842	0.217	-1.288	-0.838	-0.421	844.84	1.006
β_6	HR	-1.514	0.232	-1.979	-1.515	-1.066	923.66	1.003
σ	Residual SD	1.189	0.050	1.095	1.187	1.291	2203.22	1.000

Table 6: Posterior summaries of source-specific $\log_{10}$ concentration distributions derived from the regression model.

Source	Mean	SD	2.5%	50%	97.5%
EB (Bathing water)	3.733	0.188	3.371	3.731	4.113
ED (Drinking water)	3.120	0.110	2.894	3.121	3.333
ES (Soil)	3.889	0.105	3.683	3.889	4.095
HH (Sibling hand rinse)	3.061	0.122	2.816	3.062	3.296
HP (Mother hand rinse)	2.891	0.113	2.659	2.891	3.112
HR (Infant hand rinse)	2.219	0.140	1.934	2.223	2.489